DSC-002: Mathematical Methods

(Linear Algebra)

Lecture - 6: Quadratic Forms

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Definition: Bilinear Form

Definition

A **bilinear form** on $\mathbb{R}^n$ is a function $L = L(x, y)$ of two arguments $x, y \in \mathbb{R}^n$ which is linear in each argument, i.e. such that

1. $L(x_1 + x_2, y) = L(x_1, y) + L(x_2, y)$;
2. $L(x, y_1 + y_2) = L(x, y_1) + L(x, y_2)$.

Note (1) If $x = (x_1, ..., x_n)^T$ and $y = (y_1, ..., y_n)^T$, a bilinear form can be written as

$L(x, y) = \sum_{j,k=1}^{n} a_{jk} x_k y_j = (Ax, y) = y^* Ax$

where $\mathbf{A}_{nn}$ is determined uniquely by the linear form L.

Note (2): $L(x, y)$ is essentially an inner product.

Quadratic Forms

A quadratic form in 'n' variables $x_1, ..., x_n$ is an expression –

$$F = \sum\sum a_{kj}x_k x_j = a_{11}x_1x_1 + a_{12}x_1x_2 + ... + a_{1n}x_1x_n$$

$$. \qquad\qquad\qquad\qquad ..... \qquad\qquad\qquad\qquad$$

$$+a_{n1}x_nx_1 + ... \qquad\qquad +a_{nn}x_nx_n.$$

It is called a "quadratic form" because each term $a_{kj}x_k x_j$ contains either square of x_j or product of x_j & x_k. A QF on $\Re^n$ is the "diagonal" of a bilinear form L, i.e. any QF is: $Q[x] = L(x,x) = \langle Ax, x\rangle = x^*Ax$.

$\langle Ax, x\rangle \in \Re$ *or* C *for any* $x \in \mathbb{R}^n$ *or* $\mathbb{C}^n$. $\Rightarrow \overline{\langle Ax, x\rangle} = \langle x, Ax\rangle = \langle A^*x, x\rangle$

$\Rightarrow \langle (A - A^*)x, x\rangle = 0 \Rightarrow (A - A^*) = 0 \Rightarrow A = A^*$;

i.e. A in a QF can be made symmetric (Note: Self Adjoint transformn $A = A^*$*).*

◦ **Example**: A QF in $\mathbb{R}^2$ is : $a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2$. In matrix form -

$$\begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x^*Ax$$

Change of Variables

It is possible to simplify a QF x^*Ax by a change of variables $x = Sy$ or $y = S^{-1}x$, where S must be a nonsingular matrix. We can go either way.

- Substituting $x = Sy$ into $F = x^*Ax$ gives
$$F = (Sy)^*A(Sy) = y^*S^*ASy = y^*By, \text{ where } B = S^*AS.$$
- Note that if **A** is a symmetric matrix, **B** is also symmetric.
- The **detA** is called the **discriminant** of the QF x^*Ax.
- If $B = S^*AS$, then B is **congruent** to A.

Therefore, the discriminant of QF y^*By is: $|B| = |S^*|\ |A|\ |S| = |S|^2\ |A|$; i.e. under a nonsingular change of the variables $x = Sy$, the discriminant of new QF assumes a magnitude of $|S|^2$ times that of the original form.

- The det S is often called the **modulus of the transformation** $x = Sy$.

Definition

CONGRUENCE: A square matrix **B** is said to be congruent to the square matrix **A** if there exists a nonsingular matrix **S** such that $B = S^*AS$.

Diagonalisation of QFs

Th

By an **orthogonal transformation** *of variables every QF* x^*Ax *may be reduced to a* **diagonal form**.

Proof. Given a QF: x^*Ax, consider a nonsingular transformation of variables $x = Qy$, the matrix Q has as its columns a set of **orthonormal** eigenvectors of A which span E^n. The matrix Q is therefore an orthogonal matrix, and the transformation of variables is called an orthogonal transformation. In the **diagonal form**, the coefficient of y_j^2 is the eigenvalue λ_j of A. In terms of the variables y, the F becomes

$$y^*Q^*AQy = y^*Dy, \qquad \text{and} \qquad D = ||\lambda_j \delta_{ij}||$$

is a diagonal matrix whose diagonal elements are the eigenvalues of A.

Thus $y^*Dy = \sum_{j=1}^{n} \lambda_j y_j^2$. Only the squares of the variables appear; there are no cross products $y_i y_j$. A QF containing only the squares of the variables is said to be in diagonal form. Furthermore, we say that the transformation of variables $x = Qy$ has diagonalized the QF x^*Ax.

Orthogonal Diagonalization: Example 1:

Example. Consider the quadratic form of two variables (i.e. quadratic form on $\mathbb{R}^2$), $Q(x, y) = 2x^2 + 2y^2 + 2xy$. Let us describe the set of points $(x, y)^T \in \mathbb{R}^2$ satisfying $Q(x, y) = 1$.

The matrix A of Q is $$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Orthogonally diagonalizing this matrix we can represent it as

$$A = U \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} U^*, \quad \text{where} \quad U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix},$$

or, equivalently $$U^* A U = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} =: D.$$

The set $\{\mathbf{y} : \langle D\mathbf{y}, \mathbf{y} \rangle = 1\}$ is the **ellipse** with half-axes $1/\sqrt{3}$ and 1. Therefore the set $\{\mathbf{x} \in \mathbb{R}^2 : \langle A\mathbf{x}, \mathbf{x} \rangle = 1\}$ is the same ellipse only in the basis $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)^T$, $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)^T$, or, equivalently, the same ellipse, rotated $\pi/4$.

Orthogonal Diagonalization: Example 2

Consider the QF: $F = 2x_1^2 + 2\sqrt{2}x_1x_2 + x_2^2 = x^*Ax$

the symmetric matrix A is then $\begin{pmatrix} 2 & \sqrt{2} \\ \sqrt{2} & 1 \end{pmatrix}$

To diagonalise the form, we transform $x = Qy$, where columns of Q are - $u_1 = [-\frac{1}{\sqrt{3}}, \sqrt{\frac{2}{3}}]^T$ and $u_2 = [\sqrt{\frac{2}{3}}, \frac{1}{\sqrt{3}}]^T$ are orthonormal e.v.s of A.

Thus the transformation variables are -

$$x_1 = \frac{1}{\sqrt{3}}(-y_1 + \sqrt{2}\sqrt{y_2}), \qquad y_1 = \frac{1}{\sqrt{3}}(-x_1 + \sqrt{2}x_2)$$
$$x_2 = \frac{1}{\sqrt{3}}(\sqrt{2}y_1 + y_2), \qquad y_2 = \frac{1}{\sqrt{3}}(\sqrt{2}x_1 + x_2).$$

Note that $Q^{-1} = Q^*$; hence it is very easy to find the inverse transformation for an orthogonal transformation of variables. Since the eigenvalues of A are $\lambda_1 = 0$, $\lambda_2 = 3$, the form F becomes $F = 3y_2^2$ under this transformation of variables. The form is therefore *positive semidefinite*. The point $y = [2, 0]$ causes the form to vanish. The x corresponding to this y is $x = [\frac{2}{\sqrt{3}}, 2 - 2\sqrt{\frac{2}{3}}]$, and F vanishes for this value of x.

Theorem

If we allow x to vary over all of $\mathbf{E}^n$, then the set of values taken on by $F = x^*Ax$ is called the **Range** of the quadratic form.

Under a **nonsingular transformation** *of variables, the* **range** *of a quadratic form remains unchanged.*

Suppose, we have the form $F = x^*Ax$ *and transform it by the change of variables* $x = Sy$ *or* $y = S^{-1}x$ *to obtain the new form*

$$(Sy)^*A(Sy) = y^*S^*ASy = y^*By.$$

Now it is only necessary to note that for any x *there is a unique* y *(and, similarly, for any* y *there is a unique* x*) such that* $F = x^*Ax = y^*By$. *Hence* x^*Ax *and* y^*By *must have the same range. In general, this property will not hold if the matrix* S *is singular. For example, if* $S = 0$, *the range of* y^*By *contains only a single number* 0.

Definiteness of QFs - definitions

Positive Definite QF: The quadratic form x^*Ax is said to be positive definite if it is positive (> 0) for every x except $x = 0$.

Positive Semi-Definite QF: The quadratic form x^*Ax is said to be positive semidefinite if it is non-negative (≥ 0) for every x, and there exist points $x \neq 0$ for which $x^*Ax = 0$.

Indefinite Forms: A quadratic form x^*Ax is said to be indefinite if the form is positive for some points x and negative for others.

Negative 'definite' & 'semidefinite' forms are defined by interchanging words "negative" & "positive" in the above definitions. If x^*Ax is positive definite (semidefinite), then $x^*(-A)x$ is negative definite (semidefinite).

A symmetric matrix A is often said to be positive definite, positive semidefinite, negative definite, etc., if the respective QF: x^*Ax is positive definite, positive semidefinite, negative definite, etc.

Another set of necessary and sufficient conditions for Definiteness:

A set of necessary and sufficient conditions for the form x^*Ax to be positive definite is

$$a_{11} > 0, \quad \det\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} > 0, \quad \det\begin{pmatrix} a_{11} & a_{12} & a_{23} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} > 0, \ \ldots, \ |A| > 0;$$

There are n principal minors of A_{nn}. The QF x^*Ax is **positive definite** only if determinats of these minors are positive. → leads to convex

(Sylvesters Law) alternatively

(1) *x^*Ax is positive (negative) definite iff every eigenvalue of A is positive (negative).*

→ will lead to concave

(2) *x^*Ax is positive (negative) semidefinite iff all eigenvalues of A are non-negative (nonpositive), and at least one of the eigenvalues vanishes.*

(3) *x^*Ax is indefinite iff A has both positive and negative eigenvalues.*

Examples

(1). $F = 3x_1^2 + 5x_2^2$, $F = 2x_1^2 + 3x_2^2 + x_3^2$, $F = x_1^2$ are positive definite forms in two, three, and one variable, respectively.

(2). $F = 4x_1^2 + x_2^2 - 4x_1x_2 + 3x_3^2 = (2x_1 - x_2)^2 + 3x_3^2$ is in 3 var positive semidefinite, as it is never negative and vanishes if $x_2 = 2x_1$, $x_3 = 0$.

(3). $F = -2x_1^2 - x_2^2$, $F = -x_1^2 - x_2^2$, $F = -x_1^2$ are negative definite forms in two, two, and one variable, respectively.

(4). $F = 4x_1^2 - 3x_2^2$ is indefinite since it is positive when $x_1 = 1$, $x_2 = 1$ and negative when $x_1 = 0$, $x_2 = 1$.

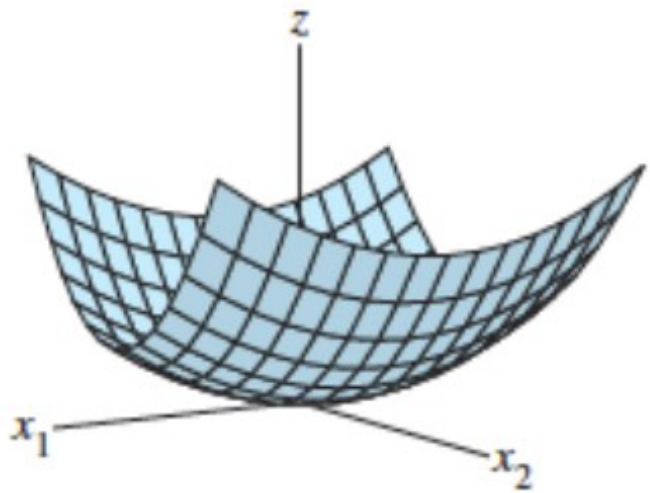
Note: *A positive (negative) definite form remains positive (negative) definite when expressed in terms of a new set of variables provided the transformation of the variables is nonsingular.*

Thus if x^*Ax is positive (negative) definite and S is nonsingular, then $(Sy)^*A(Sy) = y^*S^*ASy = y^*By$ $(x = Sy)$ is positive (negative) definite.

Q: Try to prove this statement given the previous theorem.

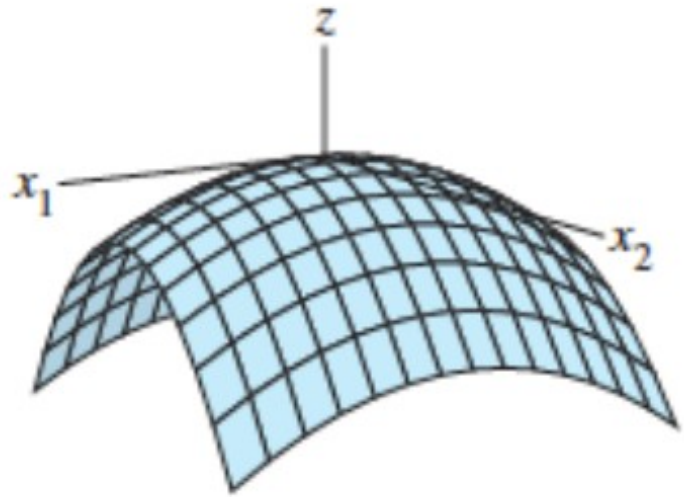
Graphs of Diagonalised of QFs

Positive definite



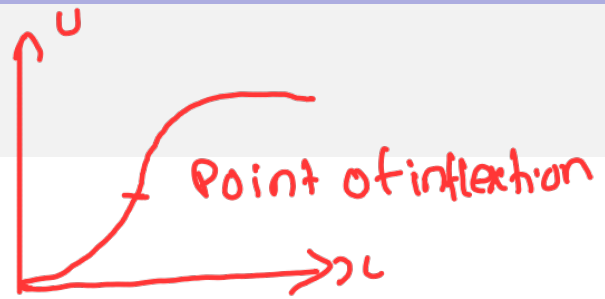
$$z = 3x_1^2 + 7x_2^2$$

Negative Definite

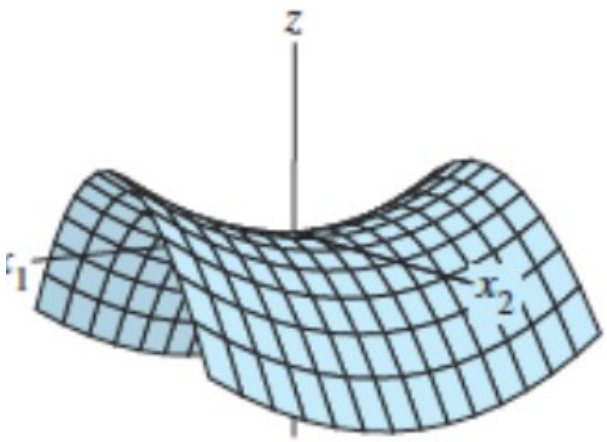


$$z = -3x_1^2 - 7x_2^2$$

Graphs of Diagonalised of QFs

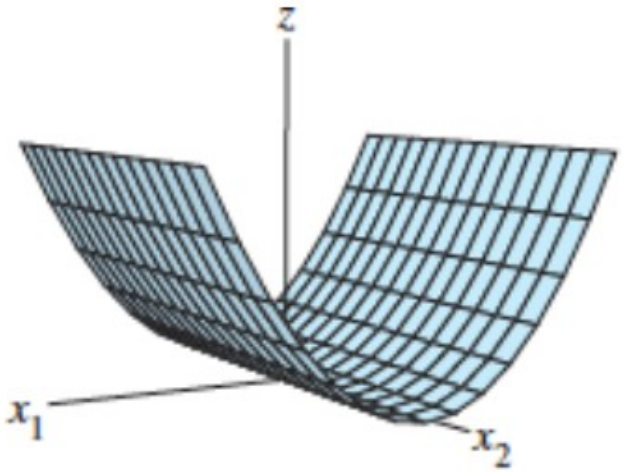


Indefinite :-
Saddle → Min Max
(Neither minimum nor maximum)



$z = 3x_1^2 - 7x_2^2$

Indefinite
Minimax, Saddle Point



$z = 3x_1^2 - 0 \cdot 2x_2^2$

z = 3 x1 ^ 2+ 0. x2 ^ 2
Then, Positive Semi-definite